

ASP.NET Core Web API

1. ASP.NET Core Web API

- **Purpose:** The core framework to build RESTful APIs without the MVC pattern.
- **Library:** Microsoft.AspNetCore.WebApi

2. Entity Framework Core

- **Purpose:** ORM for accessing the database using a code-first approach, simplifying data operations.
- **Library:** Microsoft.EntityFrameworkCore

3. Swagger / Swashbuckle

- **Purpose:** To generate API documentation for your Web API and enable interactive API exploration.
- **Library:** Swashbuckle.AspNetCore

4. AutoMapper

- **Purpose:** For mapping between DTOs (Data Transfer Objects) and entities to simplify model binding.
- **Library:** AutoMapper

5. FluentValidation

- **Purpose:** For building clean and maintainable validation logic for API models (e.g., request/response models).
- **Library:** FluentValidation.AspNetCore

6. JWT Authentication (JSON Web Token)

- **Purpose:** For handling stateless authentication using tokens.
- **Library:** Microsoft.AspNetCore.Authentication.JwtBearer

7. Cors (Cross-Origin Resource Sharing)

- **Purpose:** To configure cross-origin requests for enabling React (or other frontend apps) to make API calls.
- **Library:** Microsoft.AspNetCore.Cors

8. Serilog or NLog

- **Purpose:** For logging and monitoring API requests and errors.
- **Library:** Serilog.AspNetCore or NLog.Web.AspNetCore

9. Response Caching

- **Purpose:** Caching responses for improved performance in GET API calls.
- **Library:** Microsoft.AspNetCore.ResponseCaching

10. Background Services

- **Purpose:** For running background tasks (e.g., email sending, data processing).
- **Library:** Microsoft.Extensions.Hosting

